

NAD+

Nicotinamide Adenine Dinucleotide

Cellular Energy · DNA Repair · Longevity Pathways

What Is NAD+?

NAD+ is a coenzyme found in every living cell. It is required for life. Cells use it to produce energy, repair DNA, and regulate processes that keep tissues functioning. It cycles between NAD+ (active) and NADH (electron carrier).

How It Works

Cellular Energy NAD+ fuels mitochondria for ATP production. Low levels reduce cellular energy.	DNA Repair PARP enzymes use NAD+ to repair damaged DNA; heavy damage depletes stores quickly.
Sirtuin Activation Sirtuins need NAD+ to regulate inflammation, metabolism, stress resistance, and aging.	Metabolic Signaling Affects insulin sensitivity, fat metabolism, and circadian rhythm.

Why NAD+ Declines With Age

Levels drop ~50%+ with age due to inflammation, DNA damage, reduced NAMPT, poor sleep, alcohol, and inactivity. This weakens energy, repair, and sirtuins.

Areas of Research Interest

- Energy & fatigue (mitochondrial ATP)
- Cognitive clarity (high neuron energy demand)
- Recovery & repair (DNA maintenance)
- Metabolic health (insulin & fat metabolism)
- Healthy aging (sirtuin pathways)
- Physical performance (cellular energy)

Common Research Approaches

NMN/NR precursors boost NAD+. Direct NAD+ is also studied. Support via exercise, fasting, sleep, and low alcohol.

FOR RESEARCH PURPOSES ONLY · NOT FOR HUMAN CONSUMPTION

Educational only — not medical advice. Consult qualified professionals. All related compounds are not evaluated or approved as treatments for any condition.

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